

Molecular and Functional Characterizations of Gastrula Organizer Cells Derived from Human Embryonic Stem Cells

NADAV SHARON,^a ISHAY MOR,^a TAMAR GOLAN-LEV,^a ABRAHAM FAINSDOD,^b NISSIM BENVENISTY^a

^aStem Cell Unit, Department of Genetics, Silberman Institute of Life Sciences, The Hebrew University, Jerusalem, Israel; ^bDepartment of Developmental Biology and Cancer Research, Institute for Medical Research Israel-Canada, Faculty of Medicine, The Hebrew University, Jerusalem, Israel

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ABSTRACT

The Spemann-Mangold organizer is the structure that provides the signals, which initiate pattern formation in the developing vertebrate embryo, affecting the main body axes. Very little is known about axial induction in the gastrulating human embryo, as research is hindered by obvious ethical restrictions. Human embryonic stem cells (hESCs) are pluripotent cells derived from the pregastrula embryo that can differentiate in culture following a program similar to normal embryonic development but without pattern formation. Here, we show that in hESC-derived embryoid bodies, we can induce differentiation of cells that harbor markers and

characteristics of the gastrula-organizer. Moreover, genetic labeling of these cells enabled their purification, and the discovery of a comprehensive set of their secreted proteins, cell surface receptors, and nuclear factors characteristic of the organizer. Remarkably, transplantation of cell populations enriched for the putative human organizer into frog embryos induced a secondary axis. Our research demonstrates that the human organizer can be induced in vitro and paves the way for the study of pattern formation and the initial regulation of body axis establishment in humans. *STEM CELLS* 2011;29:600–608

Disclosure of potential conflicts of interest is found at the end of this article.

INTRODUCTION

In their milestone experiments, Spemann and Mangold [1] identified the cells at the dorsal lip of the amphibian blastopore as responsible for the patterning of the anterior-posterior and dorsal-ventral axes of the developing gastrula embryo and termed them the “organizer.” This role was realized through their ability to induce dorsal structures, and particularly neural precursors, when transplanted heterotopically. On the molecular level, the expression of several genes is characteristic of the amphibian organizer during gastrulation, the most prominent of which is the paired-type homeodomain transcription factor goosecoid (GSC). The induction of the organizer cells is regulated by the costimulation of the transforming growth factor beta (TGF β) and the WNT pathways [2]. Consequently, the organizer itself secretes numerous proteins that form intricate regulatory networks and eventually induce the TGF β pathway and inhibit the BMP and WNT pathways. Among these are Nodal related proteins [3], which are members of the TGF β family, and Dickkopf 1 (DKK1), which inhibits the WNT pathway. Cerberus 1 (CER1), another prominent marker of the organizer, is a tripartite inhibitor of all three pathways.

Numerous studies have shown high evolutionary conservation of the function and molecular basis of the organizer among vertebrates, specifically in fish, frog, chick, and mouse [4]. However, significant differences do exist. For instance, in mouse, the gastrula organizer activity seems to have been divided between two GSC expressing regions in the cup shaped early embryo, as opposed to the single structure in amphibians [5–7]. As the human gastrula differs from its murine counterpart in many aspects, the human organizer is expected to exhibit unique features. However, obvious ethical restrictions prevent the direct study of early human embryos.

Human embryonic stem cells (hESCs) are pluripotent cells derived from the pregastrula embryo [8]. In vitro, hESCs can be aggregated to form embryoid bodies (EBs) [9]. This induces their differentiation into progenitors and derivatives of the three embryonic germ layers, following a sequence similar to normal embryonic development, but without pattern formation [9–12].

Here, we study the differentiation of hESCs into cells that harbor markers and characteristics of the gastrula-organizer. Through genetic labeling of hESCs for GSC expression, we could purify and determine the gene profile of the putative

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Correspondence: Abraham Fainsod, Ph.D. Telephone: 972-2-6758157; Fax: 972-2-6415848; e-mail: fainsod@cc.huji.ac.il; or Nissim Benvenisty, M.D., Ph.D. Telephone: 972-2-6586774; Fax: 972-2-6584972; e-mail: nissimb@cc.huji.ac.il Received September 27, 2010; accepted for publication February 4, 2011; first published online in *STEM CELLS EXPRESS* February 15, 2011. © AlphaMed Press 1066-5099/2009/\$30.00/0 doi: 10.1002/stem.621

human organizer cells. Remarkably, transplantation of GSC expressing cells into frog embryos induced secondary axes, suggesting human EBs harbor organizer-like cells.

MATERIALS AND METHODS

H9 hESCs were cultured using standard procedures [10]. In vitro differentiation into EBs was performed by withdrawal of bFGF from the growth media, and factors were added with the initiation of EBs formation. ActivinA and DKK1 were purchased from PeproTech (Rocky Hill, NJ, www.peprotech.com). SB-431542 was purchased from Tocris Bioscience (Bristol, U.K., www.tocris.com) and from Cayman Chemical (Ann Arbor, MI, www.caymanchem.com). Total RNA was extracted using RNeasy Mini or Micro (Qiagen, Valencia, CA, www.qiagen.com). RNA was reverse transcribed by random hexamer priming (Promega, Madison, WI, www.promega.com) and TaqMan probes (Applied Biosystems, Warrington, U.K., www.appliedbiosystems.com) was used for real-time polymerase chain reaction (PCR). Immunostaining of cryosectioned EBs was performed using primary antibodies against GSC (Abnova, Taipei city, Taiwan, www.abnova.com), FOXA2 (Abcam, Cambridge, MA, www.abcam.com), NOGGIN (Santa Cruz Biotechnology, Santa Cruz, CA, www.scbt.com), NODAL (Santa Cruz Biotechnology, Santa Cruz, CA, www.scbt.com), CER1 (Sigma-Aldrich, Saint Louis, MO, www.sigmaaldrich.com), and β -CATENIN (Cell Signaling, Danvers, MA, www.cellsignal.com). Cy3 and Cy2 conjugated antibodies were used as secondary antibodies, and nuclear staining was performed with Hoechst 33258 (Sigma). For DiI labeling, we used Vybrant multicolor cell labeling kit (Invitrogen, Carlsbad, CA, www.invitrogen.com). Global gene expression analysis was performed using Affymetrix Gene ST1.0 microarray. Data was normalized, and genes upregulated in GSC-GFP⁺ cells were identified by being of higher expression in the GFP⁺ population in both 2- and 3-day EBs by more than 1.5 log values and by having expression above array average in the GFP⁺ cells. Dendrogram was made using the Expander integrative program suite [13]. Genetic labeling for *GSC* and *CER1* expression was made using the recombinering technique [14], and a fragment containing the sequence for eGFP and the neomycin resistance was used to replace the GSC open reading frame (ORF) in RP11-179A9 BAC or the CER1 ORF in RP11-696E8 BAC (BACPAC resources, Oakland, CA, <http://bacpac.chori.org>). The first modified BAC was linearized by digestion with AgeI (New England Bio labs, Ipswich, MA, www.neb.com) and the second with Kpn2I (Fermentas, Hanover, MD, www.fermentas.com). After restriction, the BAC was electroporated into H9 cells. Fluorescent activated cell sorting (FACS) analysis and sorting was performed using FACSCalibur and FACSria Cell-Sorting Systems (Becton Dickinson, Franklin Lakes, NJ, www.bd.com), respectively, after EB dissociation. Einsteck procedure was performed in Steinberg's solution. A small puncture was made ventrally in late blastula *Xenopus* embryos using delicate tweezers, and the chorion was removed. EBs were washed thoroughly to remove residual Activin A and dissected to expose their interior. A piece was introduced into the embryo's blastocoel through the puncture. In the refined Einsteck procedure, sorted cells were resuspended in phosphate buffered saline (PBS) with 50% Matrigel (BD Biosciences) to an estimated concentration of about 150,000 cells per μ l. Using a fine glass needle, each embryo was injected with 20–40 nl of the mixture. Treated embryos were allowed to develop further in 17°C. 5-bromo-4chloro-3-indolyl-phosphate (BCIP) or magenta phosphate was used to identify digoxigenin labeled RNA probes that hybridized to endogenous frog transcripts by in situ hybridization, as described previously [15].

RESULTS

TGF β and WNT Pathways Induce Differentiating hESCs to Express Organizer Related Genes

Numerous studies have shown high evolutionary conservation of the function and molecular basis of the organizer among vertebrates, particularly in fish, frog, chick, and mouse [4]. Similar to the *Xenopus* organizer, it seems that both TGF β and WNT pathways have a crucial role in mouse organizer formation [16–20]. To test whether these two pathways affect the establishment of the human gastrula organizer, hESCs were harvested and allowed to aggregate into EBs in the presence of Activin A and LiCl, activators of the TGF β [21] and WNT [22] pathways, respectively. mRNA was extracted 2 days after EB formation, and changes in organizer-related gene expression were analyzed using real-time PCR. Indeed, most of the genes examined were affected by the treatments, and three groups of genes could be discerned. The first group, containing genes that responded to administration of Activin A only, included *GSC*, *CER1*, *LIM homeobox 1* (*LIM1*), *hematopoietically expressed homeobox* (*HHEX*), and *NODAL* [23] (Fig. 1A). The second group, which included *BRACHYURY* [24, 25] and *CXCR4* [26], was upregulated by either Activin A or LiCl (Fig. 1B). The third group, which was represented by *CHORDIN* [27], showed no response to either treatment (Fig. 1C). As the role of WNT pathway in organizer formation is well-established, we speculated that baseline levels of endogenous WNT activity mask the activation of the pathway by exogenous factors. To test this hypothesis, we formed EBs in the presence of recombinant DKK1, a WNT inhibitor. This indeed brought about downregulation of genes from all three groups, including *GSC*, *BRACHYURY*, and *CHORDIN* (Fig. 1F). This shows that in hESCs, too, the WNT pathway is necessary for the induction of organizer related genes. Addition of the TGF β inhibitor SB-431542 similarly downregulated organizer related genes (Fig. 1F). Altogether, the effect of TGF β and WNT signaling on hESC differentiation indicates that the mechanisms underlying the induction of the human organizer are similar to those of other vertebrates.

GSC As a Marker for Putative Human Gastrula Organizer Cells

Of the genes examined, the paired-type homeodomain transcription factor GSC is expressed in the amphibian dorsal lip and in both anterior primitive streak (APS) and anterior visceral endoderm (AVE) of the mouse [5, 6, 28] and is therefore considered as a prominent marker of the gastrula organizer [29, 30]. Here, we show that many of the GSC expressing cells also express other proteins related to the organizer, such as Forkhead box A2 (FOXA2), *CER1*, *NODAL*, and *NOGGIN* (Fig. 1E). *GSC* mRNA was substantially upregulated by Activin A (more than 10-fold when compared with untreated EBs, $p = .04$), and immunostaining corroborated the effect of Activin A on GSC at the protein level (Fig. 1D). Therefore, we chose to focus on GSC as a marker for putative human gastrula organizer cells and on Activin A as an inducer of this system.

Genetic Labeling and Characterization of GSC Expressing Cells

To better characterize the GSC expressing cells, hESCs were genetically labeled to monitor GSC expression. A bacterial artificial chromosome (BAC) in which the *GSC* ORF was replaced with *enhanced green fluorescent protein* (*eGFP*) was introduced into hESCs to establish GSC-GFP hESC reporter clones (Fig. 2A). EBs from GSC-GFP hESCs treated with

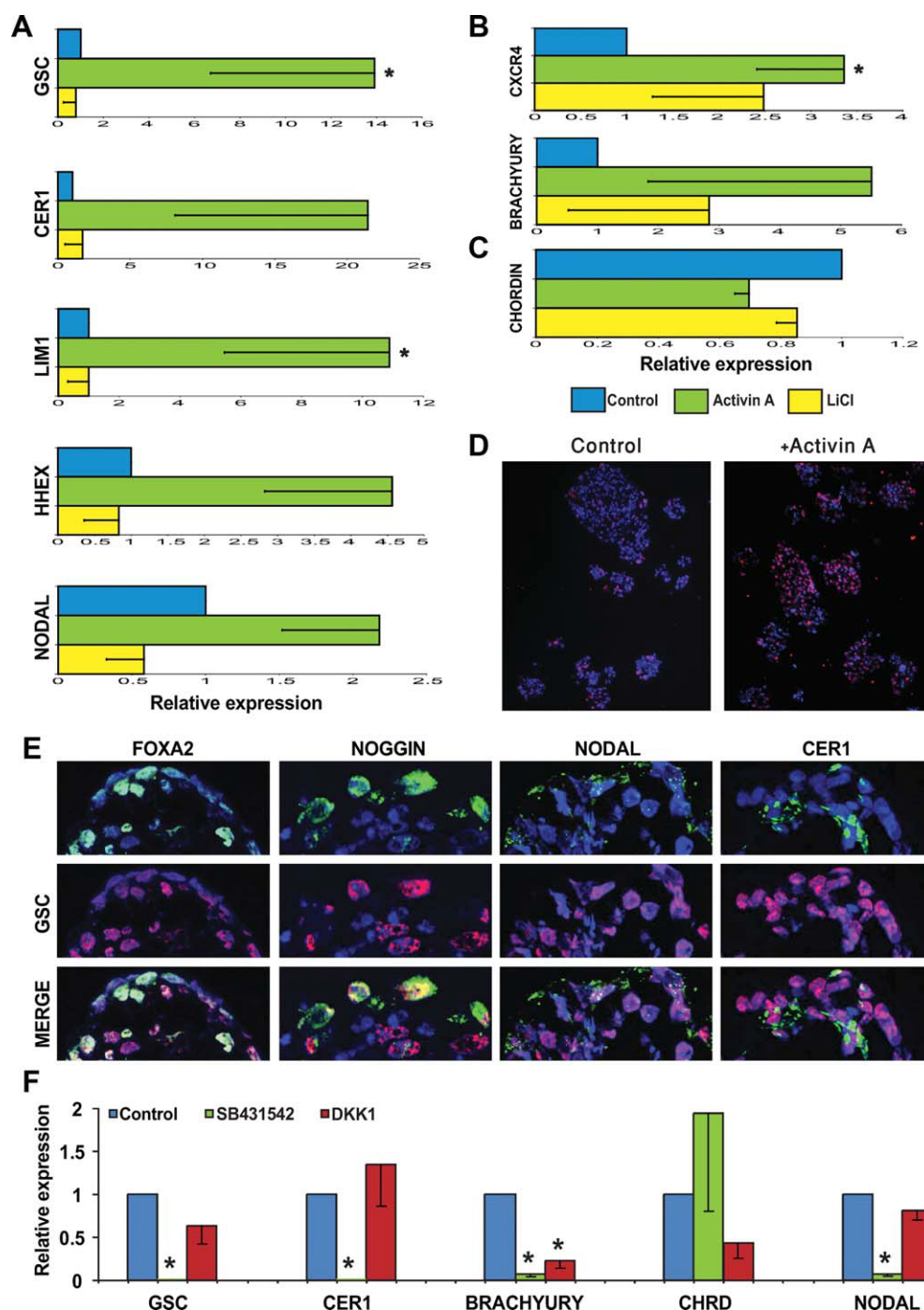


Figure 1. Induction of gastrula organizer and axis-formation related genes by aggregation of human embryonic stem cells into embryoid bodies (EBs) in the presence of Activin A and LiCl. Real-time polymerase chain reaction analysis of RNA from 2 days old EBs reveals three distinct gene expression patterns. (A): Genes that are upregulated on addition of Activin A but do not respond to addition of LiCl. (B): Genes upregulated by the addition of either Activin A or LiCl. (C): Genes that are not substantially affected by either Activin A or LiCl. Blue, control EBs; green, EBs treated with 67 ng/ml Activin A; yellow, EBs treated with 10 mM LiCl. Error bars represent minimal values as deduced from the standard error. *, $p \leq .05$, under two sided paired t test, and $n = 4$. (D, E): Immunofluorescence of sections of cryopreserved EBs. (D): GOOSECOID (GSC) protein levels are higher in Activin A treated EBs (right) versus nontreated ones (left). Original magnification, $\times 10$. (E): Many of the GSC expressing cells coexpress other organizer related proteins, such as the transcription factor FOXA2 or the secreted inhibitors NOGGIN, NODAL, and CER1. Original confocal magnification: $\times 40/1.3$. (F): Real time PCR analysis of RNA from 2 day old EBs treated with 10 nM SB-431542 or 200 ng/ml DKK1. Abbreviations: GSC, goosecoid.

Activin A contained several clusters of GFP⁺ cells (Fig. 2B) which, in contrast to mouse ESCs, did not seem to localize to a single pole within the structure [31]. Cell sorting of dissociated 2–3 days old EBs treated with Activin A showed that GSC mRNA levels were more than 7.5 times higher in GFP⁺

cells when compared with GFP[−] cells ($p < .005$; Fig. 3B), confirming the reliability of the F3 genetic labeling with the BAC construct.

FACS analysis of dissociated EBs 2 days after aggregation shows that GSC is indeed induced by Activin A (Fig. 2C). In

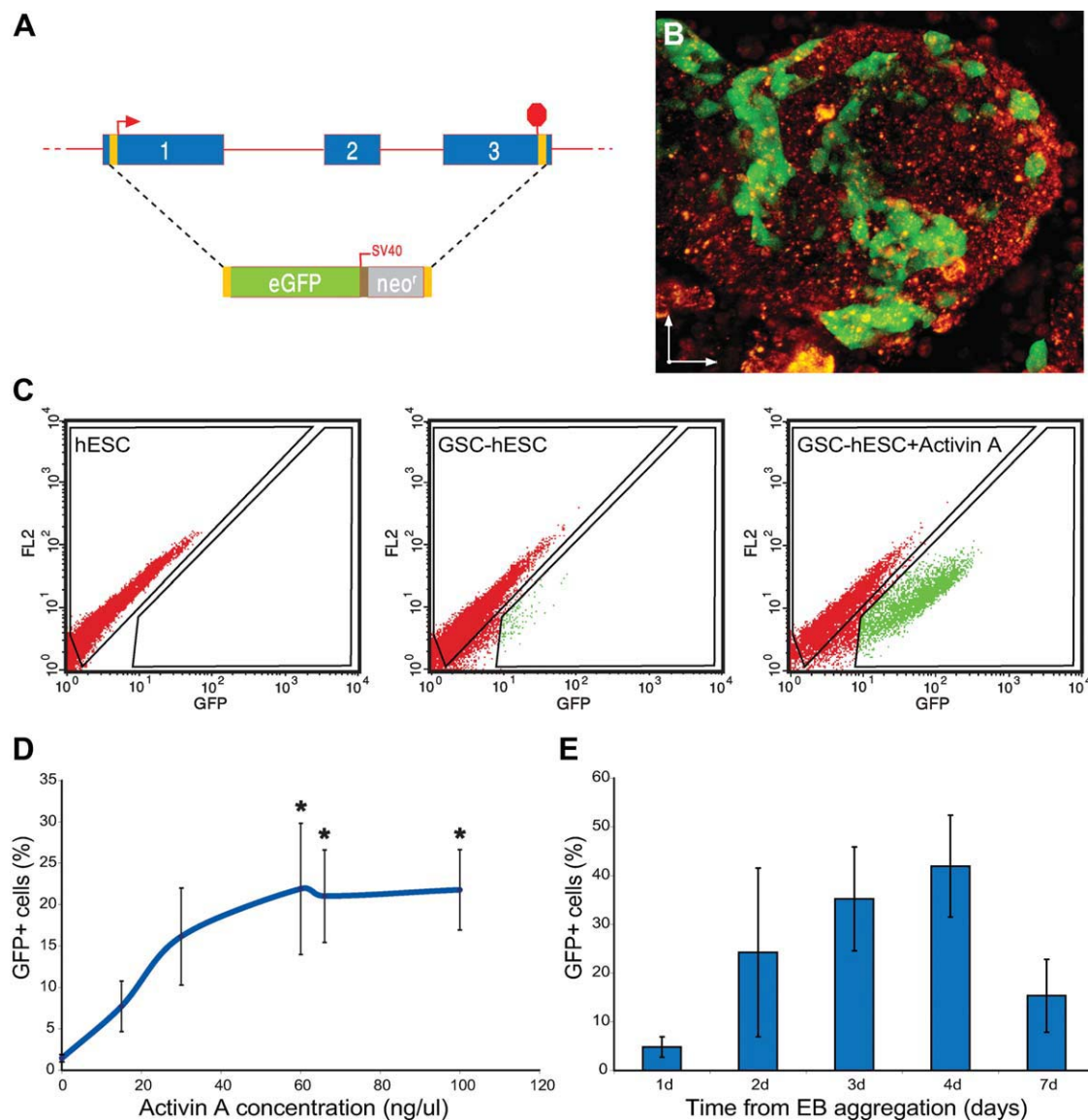


Figure 2. Genetic labeling of goosecoid (GSC) expressing cells and their characterization. (A): A scheme of a BAC containing the GSC-GFP reporter constructs. The three exons of GSC ORF were replaced with the sequence coding for eGFP adjacent to the neomycin resistance cassette under constitutive SV40 regulation. (B): The two photon microscopy imaging of a 2-day-old embryoid body (EB) made of GSC-GFP cells shows clusters of GSC expressing cells. Activin A and DiI stain were added on aggregation of the cells. Green, GSC-GFP positive cells; red, background DiI staining outlines the EB. Scale bar = 100 μ m. (C): Fluorescent activated cell sorting analysis of dissociated 2-day-old EBs. Shown from left to right: control H9 human embryonic stem cells; GSC-GFP cells; GSC-GFP cells treated with 67 ng/ml Activin A. (D): Effect of Activin A concentration on the percentage of GFP⁺ cells obtained from dissociated 2 days old GSC-GFP EBs. Error bars represent standard errors. *, $p \leq .05$, under two sided paired t test, $n = 2-5$. (E): Temporal change in the percentage of GFP⁺ cells obtained from dissociated 2 days old GSC-GFP EBs treated with 67 ng/ml Activin A. Error bars represent standard errors. Abbreviations: EB, embryoid body; eGFP, enhanced green fluorescent protein; GSC, GOOSECOID; hESC, human embryonic stem cells.

the absence of Activin A, approximately 3% of the cells showed GFP expression. On addition of Activin A at increasing concentrations, GFP⁺ cells became gradually more abundant until a plateau was reached at 60 ng/ml Activin A, with an average of approximately 20% of the cells positive for GFP (Fig. 2D). DKK1 addition abrogated Activin A induced GFP expression, suggesting that endogenous WNT activity facilitates GSC expression (Supporting Information Fig. 1B, 1C).

Activin A treated EBs started expressing GFP in a small number of cells beginning 1 day after EB aggregation, and a peak (>40%) was reached after 4 days. By the 7th day, the number of GFP⁺ cells declined dramatically (Fig. 2E). In control EBs, the number of GFP⁺ cells remained low throughout

the entire week (<3%). The transient expression of GSC in vitro corresponds to a transient population of cells, which function at the earliest stages of human gastrulation. To obtain putative gastrula organizer cells, we chose to focus on GSC-GFP⁺ cells that appear early after the induction of differentiation, that is, no more than 3 days after EB formation.

Transcriptome Analysis of GSC⁺ Cells Derived from Differentiating hESCs Reveals the Molecular Constitution of the Organizer

To elucidate the identity of the GSC expressing cells, we performed whole transcriptome microarray analysis on mRNA

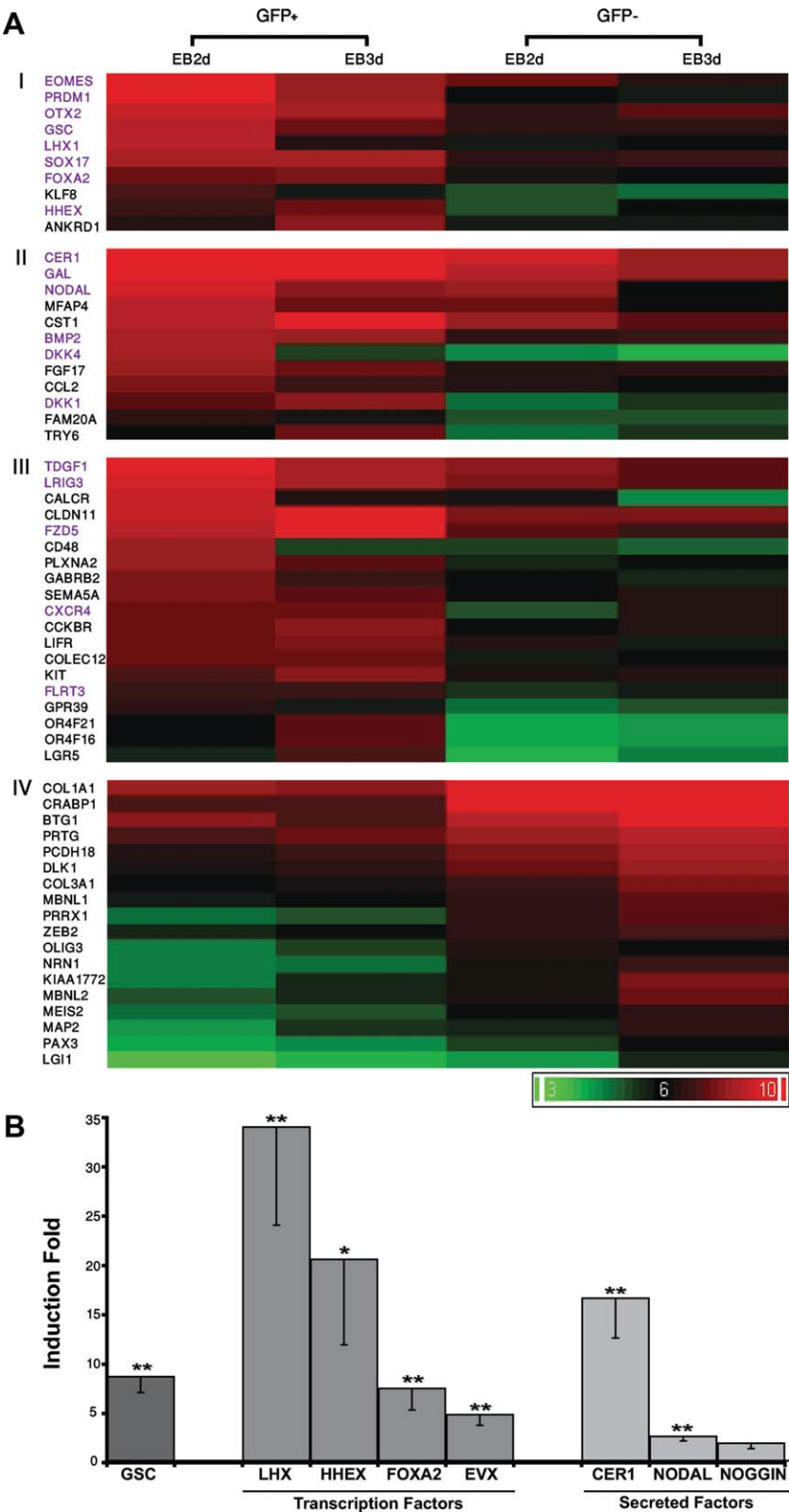


Figure 3. Genetic profiling of goosecoid (GSC)⁺ and GSC⁻ cells isolated from 2 or 3 days old embryoid bodies (EBs). GSC-GFP clones were aggregated into EBs in the presence of Activin A for 2 or 3 days. EBs were then dissociated and sorted for GSC⁺ and GSC⁻ cells using FACSARIA. (A): mRNA was extracted, and cRNA was hybridized to Affymetrix Gene ST1.0 microarrays. Data was normalized, and gene expression levels in the GSC⁺ and GSC⁻ were compared. (I) Transcription factors upregulated in the GFP⁺ cell population. (II) Secreted molecules upregulated in the GFP⁺ cell population. (III) Receptors upregulated in the GFP⁺ cell population. (IV) Genes upregulated in the GFP⁻ cell population. Names of genes previously known to relate to axis formation are in purple. (B): Real-time polymerase chain reaction analysis was performed to verify the microarray data regarding the expression of several transcription factors and secreted molecules. Error bars represent minimal values as deduced from the standard error. *, $p \leq .05$; **, $p \leq .005$, under two sided paired t test, $n = 4-8$. Abbreviations: EB, embryoid body; GFP, green fluorescent protein.

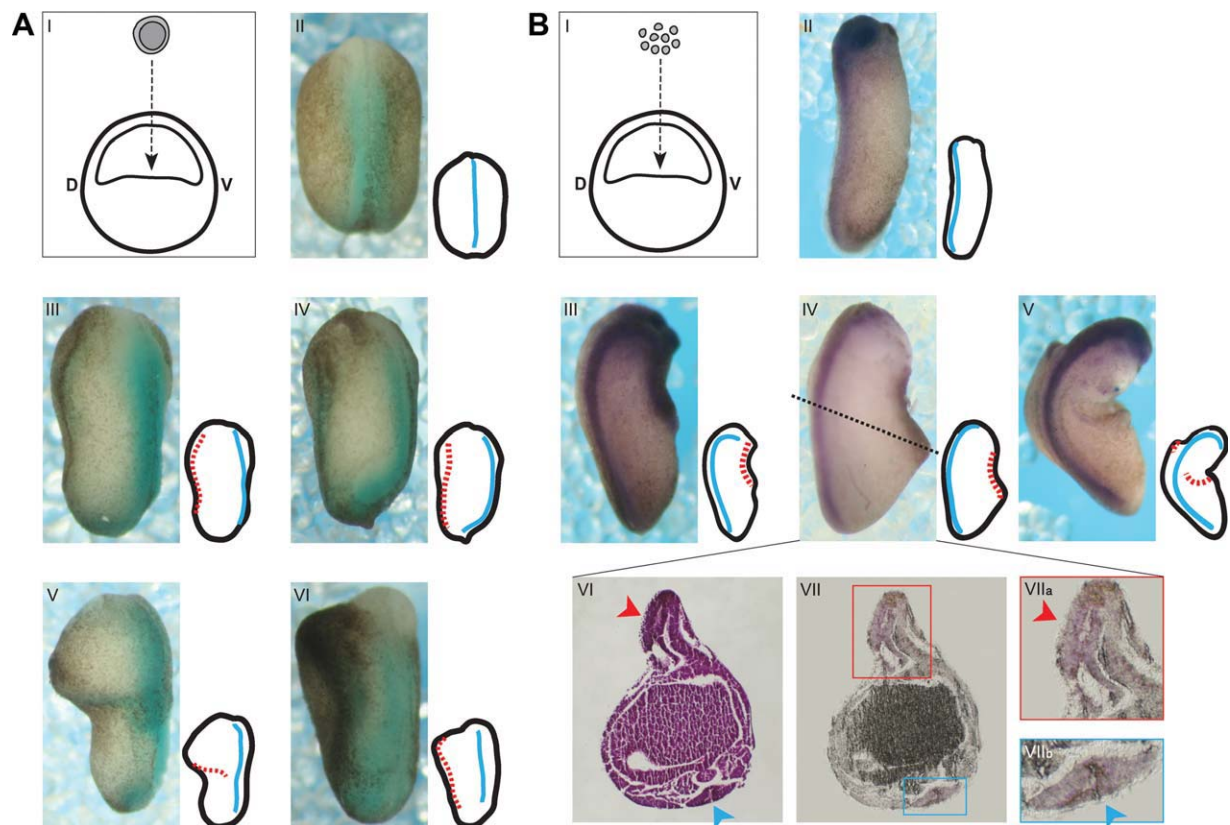


Figure 4. Transplantation of cell populations enriched for goosecoid (GSC) expression induces secondary axes. (A): (I) Outline of the Einsteck procedure. Embryoid bodies (EBs) were washed three times to remove residual Activin A and dissected using fine tweezers. A fragment was then inserted into the blastocoel of a recipient frog embryo at its ventral side (D, dorsal; V, ventral). The embryos were allowed to develop to stages 17–19 and analyzed for axis induction. (II–IV) In situ hybridization of *Xenopus* embryos using the axial markers NCAM (neural tube) and cardiac actin (somites). (II) Wild type embryo. (III–V) Embryos transplanted with 2 days old EBs treated with Activin A. (VI) Embryo transplanted with 1 day old EB treated with Activin A. (B): (I) Outline of the refined Einsteck procedure. GSC-GFP EBs were dissociated, and cells expressing GFP were separated from GFP negative cells using a cell sorter. Cells were resuspended in 50% Matrigel to an estimated concentration of more than 3,000 cells per μ l. Using a fine glass needle, the cells were injected into the ventral side of the blastocoel cavity. Embryos were allowed to develop to stages 25–26 and analyzed for axis induction. (II–V) In situ hybridization of *Xenopus* embryos using the axial marker NCAM (neural tube). (II) Wild type embryo. (III–V) Embryos injected with GFP⁺ cells, presenting conspicuous secondary axes. (VI–VII) Section through the embryo presented in IV. (VI) H&E staining outlines the primary neural tube (blue arrowhead) and the secondary neural tube induced at the site of injection (red arrowhead). (VII) Both the primary and induced neural tubes (insets at VIIb and VIIa, respectively) are stained for NCAM. For each figure, a diagram of the embryo is shown. Blue line represents primary dorsal axis and red dashed line represents induced secondary axis. Abbreviations: D dorsal; V, ventral.

extracted from sorted GFP⁺ and GFP[−] cells isolated 2 or 3 days after EB formation, in the presence of Activin A. Seventy-five genes were found to have substantially higher expression levels in the GSC-GFP⁺ cell population, and for several transcription factors and secreted proteins, this was verified by real-time PCR (Fig. 3B). We compared the expression of genes enriched in the GSC-GFP⁺ cell population with the expression of their frog homologs in the dorsal side of stage 10.5 *Xenopus* embryos [32]. This analysis showed that both the hESC derived GSC-GFP⁺ cells and the frog dorsal region express the classical organizer markers (*GSC*, *DKK1*, and *CER1*), alongside other organizer related genes (Supporting Information Fig. 2). Furthermore, analysis of available literature revealed that the genes enriched in the GSC-GFP⁺ cell population show an overwhelming enrichment of organizer-related genes. Nine of eleven (82%) transcription factors (Fig. 3A, I) are known from other vertebrates to have a role related with organizer function or to be expressed in the organizing regions. These include *GSC* itself, *FOXA2*, *MIXL1*, *LIM1*, *SOX17*, *EOMES*, and others. *EOMES* is also known to be involved in epithelial to mesenchymal transition,

a hallmark of organizer function, and indeed the GSC-GFP⁺ cell population showed higher *N-CADHERIN* expression and lower *E-CADHERIN* levels when compared with the GFP[−] cells obtained from 2 days old EBs. Among the secreted molecules strongly coexpressed with *GSC* in the GFP⁺ cells (Fig. 3A, II), 5 of 12 (41%) are organizer-related and so are 5 of 19 (26%) receptors (Fig. 3A, III). The *Xenopus* organizer is known to secrete inhibitors of both the WNT and the BMP pathways, in parallel to activation of the Nodal/Activin pathway. Indeed, the genes upregulated in the GSC-GFP⁺ cells include two members of the DKK family of WNT inhibitors (*DKK1* and *DKK4*), alongside with *CER1*, a tripartite inhibitor of WNT, BMP, and TGF β [33]. Also upregulated in the GSC-GFP⁺ cells are *NODAL*, the Nodal/Activin activator, and its coreceptor, *teratocarcinoma-derived growth factor 1* (*TDGF1*). Although it activates the competing BMP pathway, we found *BMP2* to be upregulated in the GSC-GFP⁺ cells. This may correlate with its expression, together with *anti-dorsalizing morphogenic protein* (*ADMP*), at the *Xenopus* organizer, where it seems to have a role in limiting organizer expansion [34–36].

The genes upregulated in the GSC-GFP⁺ cells represent a more heterogeneous population, as they cannot be attributed to any particular known cell type (Fig. 3A, IV). Follistatin is the only organizer related molecule enriched in them. However, during early frog development, Follistatin seems to be excluded from the GSC expressing cells within the organizer [37], further emphasizing the striking resemblance between amphibian and human GSC⁺ cells. Gene ontology based functional annotation [38] reveals a significant enrichment of genes related to neural development within the GSC-GFP⁺ cells, including *CRABP1*, *DLK1*, *OLIG3*, *PAX3*, and others. As neural induction is a hallmark of organizer function, it is possible that the GSC expressing cells had induced this fate on neighboring hESCs during their initial differentiation.

To further verify the molecular identity of the GSC⁺ cells, we established additional lines of hESCs that were now labeled for expression of *CER1*, a secreted molecule related with the organizer (Supporting Information Fig. 3). Although not identical to GSC⁺ cells, the CER1⁺ cells show very high similarity with their molecular composition. As GSC expression is probably shared by a few cell populations, this similarity shows that the subpopulations are highly similar, and most probably differ in the expression of only a low number of genes.

Transplantation of GSC-GFP⁺ Cells into Frog Gastrula Induces a Secondary Axis

After demonstrating that hESCs can be induced to differentiate into cells with the molecular signature of the gastrula organizer, we wanted to see if the same culture conditions could also establish the organizer function. Previously, Blum et al. [29] showed that transplantation of the GSC expressing distal tip of gastrula stage mouse embryos into *Xenopus* embryos induced partial secondary dorsal axes—thus identifying the tip as the area containing the murine gastrula organizer. To demonstrate a human gastrula organizer function, differentiated hESCs enriched for GSC expressing cells were transplanted into late blastula (st. 8–8.5) *Xenopus* embryos using the Einsteck procedure [39]. A small puncture was made in the ventral animal cap, and a fragment from an Activin A treated EB was first thoroughly washed to remove residues of the factor, and then inserted through the incision into the blastocoel cavity (Fig. 4A, I). The embryos were allowed to develop for an additional 24–36 hours. In situ hybridization was performed with probes specific for various frog axial markers including *NCAM* (neural tube). Transplantation of early (1–2 days old) EBs treated with Activin A induced a secondary axis in more than 25% of the transplanted embryos (15 of 57) (Fig. 4A, II–VI; Supporting Information Table 1A). If the transplanted EB was grown in the absence of Activin A, the frequency of axial structure induction was significantly lower (5 of 51, $p = .045$, Fisher's exact test), similar to the results obtained from sham operated embryos (3 of 46, 6.5%). The latter is probably caused by scarring inflicted by the intrusive Einsteck procedure. This, we assume, may also be the cause for some of the locally restricted patches of staining observed in several cases after in situ hybridization.

To better investigate the organizing activity of the GSC expressing cells within the EBs, we chose to refine the Einsteck procedure (Fig. 4B, I). For that purpose, GSC-GFP genetically labeled cells were aggregated into EBs in the presence of Activin A and allowed to differentiate for 2 days. The EBs were dissociated and either GFP⁺ or GFP[−] sorted cells were injected through a fine needle into the blastocoel cavity of *Xenopus* embryos (approximately 3,000 cells per embryo). On reaching st. 25–26, the embryos were examined for the induction of secondary axial structures by in situ hybridization with a probe specific to the frog

NCAM (Fig. 4B, II–VII). Embryos injected with GFP⁺ cells contained an induced secondary axis in 22% of the cases (11 of 48 and Supporting Information Table 1B), whereas the GFP[−] injected embryos showed axial inductions in less than 5% of the embryos (2 of 45), a difference which proved statistically significant ($p < .015$, Fisher's exact test, two independent experiments). Notably, many of the structures induced by the GFP⁺ cells were composed of clear elongated axes, none of which was present in the embryos injected with the GFP[−] cells. Histological examination showed that the GFP⁺-cells could induce an axis in which the epithelial cells fold to form the typical tubular morphology of a neural tube (Fig. 4B, VI, VII).

DISCUSSION

Gastrulation involves both the differentiation into the three embryonic germ layers and the patterning of the main body axes. Previous studies in hESCs have related mostly to the first aspect and regarded GSC as one of several markers for definitive endoderm [40, 41]. Here, for the first time, we isolate human GSC expressing cells formed during the earliest stages of hESCs differentiation and examine their role as the putative human organizer. We show that these cells are induced by molecular pathways known to induce the organizer in other vertebrates, and that they express genes related to the establishment of the early embryonic axes. Finally, we show that these cells possess the function of the gastrula organizer, as they can induce the formation of a neural tube when transplanted ventrally to blastula stage frog embryos.

The effect of TGF β and WNT signaling on hESC differentiation indicates that the mechanisms underlying the induction of the human organizer are similar to those of other vertebrates. The amphibian organizer can be subdivided into more specialized organizers which pattern the head, the trunk, and the tail [42]. In mouse embryos, the gastrula organizer activity seems to have been divided between two GSC expressing regions in the cup shaped early embryo (approximately 6.5 days post coitum). In the posterior part of the embryo, the organizing center is the APS [5] that contains mesendoderm and is the physiological equivalent of the amphibian dorsal lip of the blastopore. In the embryo's anterior region, the extra embryonic cells at the AVE behave as an organizing center [6, 7]. Accordingly, our results point to the fact that the GSC⁺ cell population is heterogeneous (Fig. 1). However, as observed by the similarity between GSC⁺ and CER1⁺ cells, these subpopulations are expected to differ only slightly. Future work should focus on identifying the components of the human organizer, and intraspecies transplantations can provide means to examine their distinct roles.

A comprehensive transcriptome analysis of isolated GSC⁺ cells revealed that many organizer related genes are coexpressed with GSC. The varying proportion of known organizer related genes among different functional groups observed in genes upregulated in GSC⁺ cells conforms with the current knowledge regarding axis formation. Most of our information is of transcription factors involved in the process, as these stand at the top of the process, and easily show a phenotype when manipulated. Similarly, as the organizer secretes strong inducers to perform its function [43], many of its secreted molecules are known. However, little is known about the receptors on the surface of organizer cells. These receptors are assumed to be involved, among other things, in the regulation of organizer localization and in directing cell migration during axis formation. Indeed, the receptors upregulated in GSC⁺ cells include, among others, *PLXNA2* [44] and

SEMA5A [45], known to take part in cellular position determination during axonal guidance. The genes upregulated in the GSC expressing cells may, therefore, possess a yet unknown role in the patterning of the mammalian and particularly the human axis.

When injected ventrally into frog blastula-stage embryos, GSC^+ cells induced differentiation of a secondary neural tube. Although the axes induced by direct injection of GSC^+ cells seem more complete than those obtained by EB transplantation, they are still partial in comparison to those reported when frogs are transplanted with organizers of their own species. However, it should be noted that in interspecies transplantations, most secondary axes are incomplete as in the case of the mouse distal tip [29]. We assume that the suboptimal conditions for hESCs growth in the frog embryo hampered their further proliferation and differentiation and thus may have also reduced their axis inducing ability. However, the induction of host cells to possess molecular and morphological hallmarks of a neural tube clearly demonstrates that the human GSC^+ cells act as a gastrula organizer. Functional manipulation of either the host embryo or the transplanted human organizer cells (using Morpholinos or small molecules) should allow further understanding of the molecular interactions, which facilitate the organizer.

Understanding the mechanisms regulating early shape induction in the embryo is important for both the ability to promote organ differentiation in vitro and for deciphering early human embryogenesis. For the first time, we demonstrate that hESCs can differentiate into cells with the molecular signature and function of the gastrula organizer and present an experimental model-system that should allow the study of early body plan patterning in the human embryo. Molecular analysis of GSC^+ cells and their ability to induce a secondary axis in frog embryos emphasize the extraordinary evolutionary conservation in organizer function between human and other vertebrates. Future work should focus on characterizing the subpopulations

within the GSC^+ cells and on the roles of the genes, we identified as organizer-related, in axis formation.

CONCLUSION

Human embryoid bodies derived from embryonic stem cells express markers characteristic of the gastrula-organizer. Genetic labeling of the organizer cells enabled their purification and molecular characterization. Interestingly, transplantation of these cells into frog embryos induced a secondary axis. The discovery of the human gastrula organizer within human embryoid bodies paves the way for the study of pattern formation and body axis establishment in humans.

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DISCLOSURE OF POTENTIAL CONFLICTS OF INTEREST

The authors indicate no potential conflicts of interest.

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